

Latent Planning for Long-Horizon Event Prediction

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Visual overview

Long-horizon prediction requires a model to preserve useful information and decide how far into the future to reason. The Predictron learns a latent Markov reward process and blends predictions from several internally imagined planning depths.

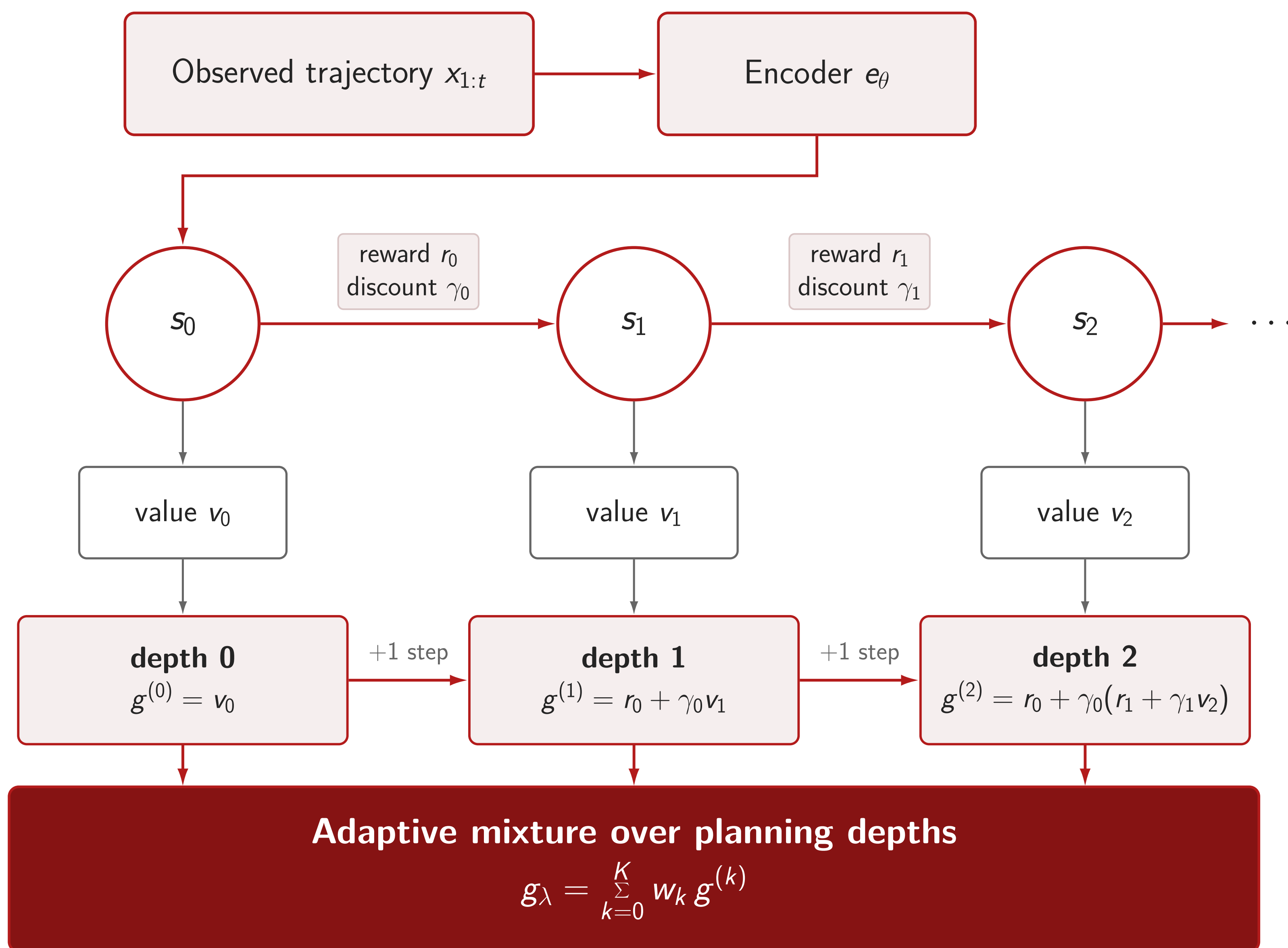


Figure 1: Latent planning architecture. Adapted from Silver, D., Hasselt, H., Hessel, M., Schaul, T., Guez, A., Harley, T., & Degris, T. (2017, July). The predictron: End-to-end learning and planning. In *International Conference on Machine Learning* (pp. 3191–3199). PMLR.

Predictron

k -Step Pretun:

$$g^k = r^1 + \gamma^1 (r^2 + \gamma^2 (\dots + \gamma^{k-1} (r^k + \gamma^k v^k) \dots)) \quad (1)$$

λ -Pretun:

$$g^{\lambda} = \sum_{k=0}^K w_k g^k \quad (2)$$

where:

$$w^k = \begin{cases} (1 - \lambda^k) \prod_{j=0}^{k-1} \lambda^j & \text{if } k < K \\ \prod_{j=0}^{K-1} \lambda^j & \text{if } k = K \end{cases} \quad (3)$$

Backward Accumulation Sweep:

$$g^{k,\lambda} = (1 - \lambda^k) v^k + \lambda^k (r^{k+1} + \gamma^{k+1} g^{k+1,\lambda}) \quad (4)$$

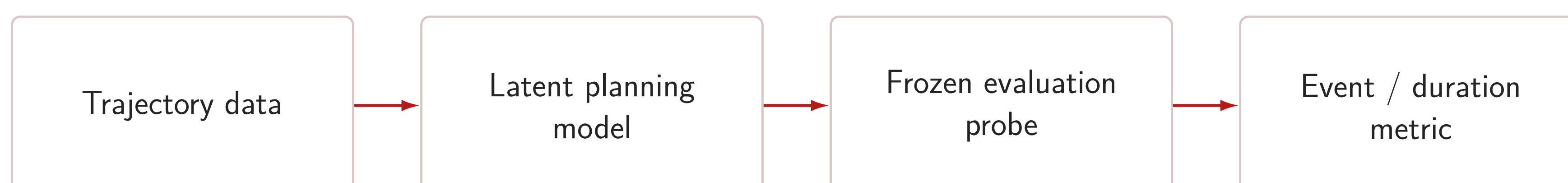
Loss function

$$L^{0:K} = \frac{1}{2K} \sum_{k=0}^K \|\mathbb{E}_p[g | s] - \mathbb{E}_m[g^k | s]\|^2 \quad (5)$$

$$L^{\lambda} = \frac{1}{2} \|\mathbb{E}_p[g | s] - \mathbb{E}_m[g^{\lambda} | s]\|^2 \quad (6)$$

$$L_{\text{cons}} = \frac{1}{2} \sum_{k=0}^K \|\mathbb{E}_m[g^{\lambda} | s] - \mathbb{E}_m[g^k | s]\|^2 \quad (7)$$

Experimental pipeline



Introduction

Motivate the problem in 3–5 sentences. State why it matters and what gap your work fills.

- Key point one
- Key point two

Markov reward process (MRP)

An MRP is defined by:

$$s', r, \gamma = p(s, \alpha) \quad (8)$$

- s' : Next state
- r : Reward
- γ : Discount factor

Value function of an MRP p : the expected return from state s

$$v_p(s) = E_p[g_t | s_t = s] \quad (9)$$

$$v_p(s) = E_p[r + \gamma v_p(s') | s] \quad (10)$$

Equations

$$\hat{y} = \arg \max_y p(y | x; \theta) \quad (11)$$

Results

Method	Accuracy	F1
Baseline	0.812	0.795
Ours	0.904	0.891

Table 1: Comparison on the test set.

Conclusions

- Main takeaway
- Limitation and future work

References

[1] Silver, D., Hasselt, H., Hessel, M., Schaul, T., Guez, A., Harley, T., & Degris, T. (2017, July). The predictron: End-to-end learning and planning. In *International Conference on Machine Learning* (pp. 3191–3199). PMLR.